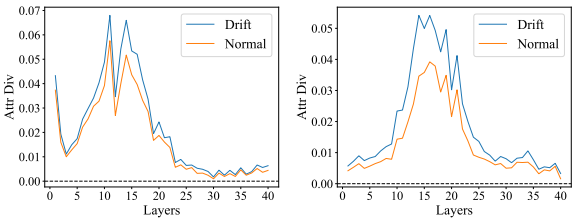
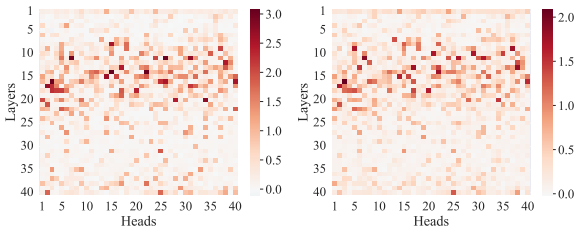


(a) Llama2-13B (b) Baichuan2-13B

Figure 3: Information flow divergence comparison on Winogrande.



(a) Winogrande (b) CSQA

Figure 4: Attention tracing results across different attention heads on Llama2-13B.

CoT performance. For datasets, we select Winogrande (Sakaguchi et al., 2020) and CSQA (Talmor et al., 2019).[4] The detailed implementation of this experiment is shown in Appendix C.

Result and Analysis Figure 2 illustrates our experimental results on Winogrande (The results on CSQA are shown in Appendix C). We can find that: **(Claim 1) When the Rationale Drift issue occurs, CoT receives less information from the question context compared to the correct case.** On both datasets and different models, there is a significantly lower information flow between the question context and CoT when the LLM generates a drifting rationale (the blue line) compared to the correct one (the orange line). This aligns with Hypothesis 1. **(Claim 2) The shallow attention layers are crucial for LLMs to extract contextual information.** In all cases, both the information flow and the gap peak at around the 15th attention layer, indicating these layers are significant sites for the rationale generation.

Supplementary Experiment In the main experiment, we use golden labels to generate the correct CoT. To eliminate the influence of this additional factor on our results, we design a supplementary experiment. Concretely, we first use the label to generate CoTs from correct reasoning cases, which serves as a control group. Then, we compute the context information divergence between the newly generated CoT c_n and the original one c_o, i.e.:

$$Attr(c_n|c_o) = Q^{(l)}_{qc_n} - Q^{(l)}_{qc_o} \tag{3}$$

where q is the question context. We compare this divergence between the control and drifting group, whose results are reported in Figure 3 and Figure 10. As we can see, the gap between the correct CoT and the drifting one (the blue line) is larger than the control group (the orange line) and the max divergence occurs in shallow layers (around the 15th layer). **This indicates that a correct CoT indeed gets more information flow from the context in shallow attention layers, validating the effectiveness of Claim 1 and 2.**

3.3 Attention Tracing Experiment

Experimental Settings We also design an experiment based on attention scores to validate Hypothesis 1 from another perspective. For a pair of rationales <c, c^*> targeting the same question context q (c denotes the correct CoT and c^* is the drifting one), we compute their attention divergence:

$$Attn(c|c^*) = \sum_{(i,j)\in C_{qc}} \frac{A^{(l,h)}_{i,j}}{|c|} - \sum_{(i,j)\in C_{qc^*}} \frac{A^{(l,h)}_{i,j}}{|c^*|} \tag{4}$$

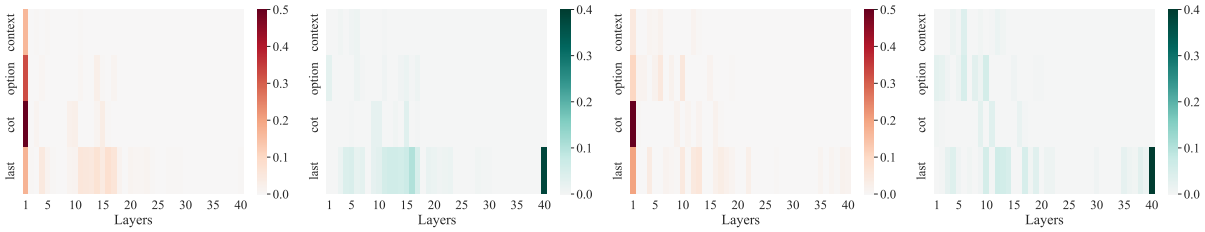
Here, we replace the $Attr(A^{(l)}_{i,j})$ in Equation 2 with the weights on the h-th attention matrix head $A^{(l,h)}_{i,j}$ and repeat the calculation in Equation 3.

Result and Analysis The results on two datasets are shown in Figure 4 and Figure 11. We can get the following observations: (1) The attention divergence is greater than 0 in most heads, which indicates **a lack of information from the question context in Rationale Drift cases (consistent with Claim 1).** (2) **The largest attention divergence appears around layer 15, which is consistent with the sites we find in Claim 2.** This once again illustrates that attention heads of these layers are crucial for the LLM to obtain contextual information when generating CoT.

4 Tracing Information Flow in Answer

In this section, our goal is to verify the information loss based on Hypothesis 2. To achieve this, we first introduce the main tracing method in this section, which is called the causal tracing method

[4]Unless otherwise specified, we use the same models and datasets in the following probing experiments.



(a) Correct Case's Attn (b) Correct Case's MLP (c) Drifting Case's Attn (d) Drifting Case's MLP

Figure 5: Intervention tracing results on Winogrande in correct and drifting answering cases.

(§4.1). Next, by employing it, we trace the information flow in the answer generation stage and identify the mechanism behind the Answer Drift issue through comparative experiments (§4.2). At last, we apply the attribution tracing method to verify our hypothesis from another perspective (§4.3).

4.1 Tracing Method

Since the task we study is in the form of multiple-choice questions, we set our focus on the feedforward pass that predicts the label. Inspired by the previous works (Meng et al., 2022; Stolfo et al., 2023; Geva et al., 2023), we take the causal tracing method to quantify the contribution of different intermediate variables during this pass. Specifically, for hidden states $h_i^{(l)}$ in a clean run that predicts the answer, we have:

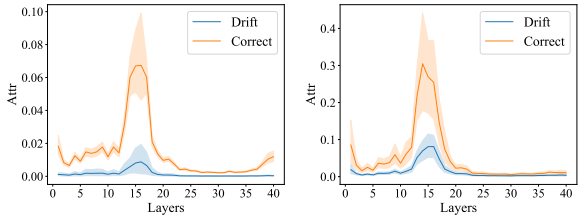
$$\begin{aligned} h_i^{(l)} &= h_i^{(l-1)} + a_i^{(l)} + m_i^{(l)} \\ a_i^{(l)} &= attn^{(l)}(h_1^{(l-1)}, ..., h_i^{(l-1)}) \\ m_i^{(l)} &= mlp^{(l)}(a_i^{(l)} + h_i^{(l-1)}) \end{aligned} \tag{5}$$

where i, l is the i-th token in the l-th layer, $a_i^{(l)}, m_i^{(l)}$ represents the activations of attention and MLP modules in Transformer (Vaswani et al., 2017). Supposing that a certain input part is represented as $z = [v_i^{(l)}, ..., v_j^{(l)}]$ after passing through a model component, we set $v_k^{(l)} = v_k^{(l)} + \epsilon$ for $k \in [i, j]$ to intervene this hidden vector, where ϵ is Gaussian noise.[5] Thus, we can compute the direct effect (DE) of this component:

$$DE(z) = \frac{P(o) - P_z^*(o)}{P(o)} \tag{6}$$

where $P(o)$ is the probability of the model's final prediction, $P_z^*(o)$ is the probability after the intervention. Therefore, through this metric, we can quantify the contributions of different components in changing the final prediction, thereby tracing the information flow in this stage.

[5]We select ϵ to be 3 times larger than the empirical standard deviation of hidden embeddings in each dataset.



(a) Winogrande (b) CSQA

Figure 6: Attribution tracing results on Llama2-13B during the answer generation stage.

4.2 Intervention Tracing Experiment

Experimental Settings We sample correct and drifting answering cases from datasets, average over them and compute the average direct effect (ADE). Here we compute the impact of four components on the final prediction: context (question contexts), option (question options), CoT, and last (the last token before the label prediction).

Results and Analysis We report the result of Llama2-13B on Winogrande in Figure 5 and others in Appendix D, from which we can get two conclusions: **(Claim 3) For attention modules, drifting cases loss information from the question.** When the answer is correct, we can observe a high effect on the context and option in the first layer (see Figure 5a). But for the drifting case, the LLM extracts limited information at these positions (see Figure 5c). This aligns with Hypothesis 2. **(Claim 4) For MLP modules, the information is not lost.** We observe the same high-effect sites in the last layer and shallow layers of the last token, they do not show regular differences (see Figure 5b and 5d).

4.3 Attribution Tracing Experiment

Experimental Settings For further validation of our hypothesis, we also use the attribution score in §3 to trace the information flow in this stage. Referring to Equation 2, we compute the score between the question context and the last token

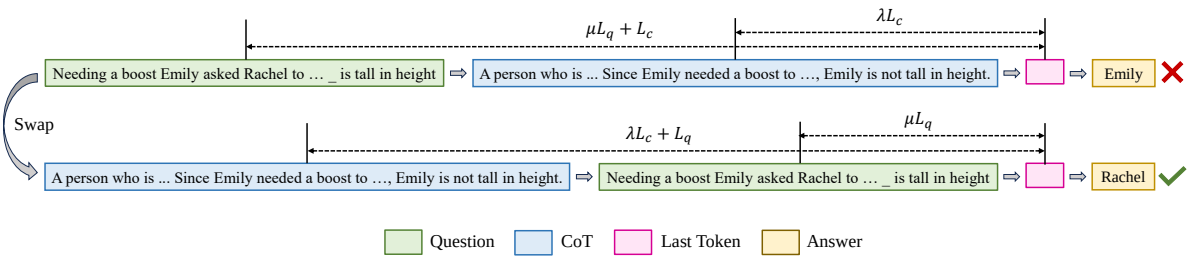


Figure 7: An example of our serial-position swap method.

(since it's used for generating the answer). We set the $F(\cdot)$ in Equation 1 as the loss for predicting the final answer, comparing the scores for correct and drifting cases after averaging across samples.

Results and Analysis The results of this experiment are reported in Figure 6 and 15. We can observe that, **when the Answer Drift issue occurs, the information flow from the question significantly decreases**. This verifies the information loss we mention in Hypothesis 2 and Claim 3.

5 Mitigating Toxic CoT Problem

In this section, we propose a novel method called RIDERS (**R**esidual decod**I**ng and s**ER**ial-position **S**wap) to address the Toxic CoT problem. We first introduce the two components in it, which are designed based on Hypothesis 1 and 2, respectively (§5.1). Then, we conduct experiments on commonsense reasoning benchmarks, demonstrating the effectiveness of our method (§5.2 and §5.3). At last, we conduct extra experiments to further emphasize the contribution of our approach (§5.4).

5.1 Mitigation Method

Residual Decoding We design a new decoding methodology to address the Rationale Drift issue, in which we construct a virtual residual structure during the CoT generation, "connecting" the question context with each CoT token. Our decoding algorithm is demonstrated in Algorithm 1. In each iteration of generating a new token, we first select the top n tokens with the highest probabilities and record their logits scores (line 3). Then we calculate the attention score between the context and current token like Equation 4, normalize it, and add it as an additional reward to promote more information flow (lines 6,7). Finally, we select the token with the highest score to update the input and repeat the process until the termination condition is met. We use the attention matrix in layer 15 to compute the attention score, since it is crucial for the exchange of contextual information according to Claim 2. More implementation details of this method are provided in Appendix E.

Algorithm 1 Residual Decoding Algorithm

Require: model $\mathcal{M}$, input x, question context q, candidate_num n, weight ω.

1: **for** iteration $i \in 0, 1, ...$ **do**
2: $\quad logits = \mathcal{M}(x)$
3: $\quad tokens, scores = \text{top}(logits, n)$
4: $\quad$ **for** $j \in 1, ...n$ **do**
5: $\quad\quad t = tokens[i]$
6: $\quad\quad attn_score = \text{Attn}(q, t) \,/\, \text{Attn}(x, t)$
7: $\quad\quad scores[j] = scores[j] + \omega * attn_score$
8: $\quad$ **end for**
9: $\quad idx = \text{argmax}(scores)$
10: $\quad t = tokens[idx]$
11: $\quad x = x + t$
12: $\quad$ **if** stop(t) **then**
13: $\quad\quad$ **break**
14: $\quad$ **end if**
15: **end for**
16: **return** x

Serial-Position Swap In this method, we attempt to compensate for the information lack in the Answer Drift issue. According to previous research on the serial-position effect in context, models tend to utilize information better at the beginning and end of the input (Qin et al., 2023; Liu et al., 2023b). In our work, the beginning of the input are prompts, while the current question and the generated CoT are both located at the end. Therefore, when they are closer to the last token, their information is more easily utilized in the final prediction. As in Figure 7, we denote the lengths of the question and CoT as L_q and L_c, and assume that the key information is located at positions μL_q and λL_c (similar to the center of mass in physics). We can infer that, after the swapping operation in Figure 7, the distance from the question to the end is reduced ($\mu L_q + L_c \rightarrow \mu L_q$). Besides, if we consider the total distance from the question and CoT to the end, we can perform the following calculation: